

Miraflor Santos

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EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY WOODS HOLE OCEANOGRAPHIC INSTITUTION

Doctorate in Biological Oceanography

Coursework: Applied Statistics, Ecology, Scientific Machine Learning

Boston and Woods Hole, MA
Aug 2019 - 2025

NEW YORK UNIVERSITY, ABU DHABI (NYUAD)

B.S. (Hons.) in Biology (GPA: 3.78/4)

Abu Dhabi, UAE
Aug 2014 - May 2019

EXPERIENCE

SOSIK LAB

Graduate Assistant

- Developing metric for measuring and predicting resilience of open-ocean food web on the Northeast U.S. Shelf
- Developing statistical models for linking environmental factors to phytoplankton abundance on the Northeast U.S. Shelf
- Developed method to detect heterotrophic bacteria from decadal time series using staining with the Attune NxT Flow Cytometer
- Implemented DADA2 algorithm for taxonomic classification of genomic microbial time series

Woods Hole, MA
Aug 2019 - Present

FRONTIER DEVELOPMENT LAB

Researcher

- Developed Neural Radiance Field model to generate 3-D map of the Sun from EUV images
- Developed machine learning model to predict solar irradiance from EUV images

Mountain View, CA
May 2022 - Aug 2022

AMIN LAB

Research Assistant

- Worked with **Prof. Shady Amin** on automated analyses of genetic, transcriptomic, and metabolomic data from co-culturing experiments with phytoplankton and bacteria
- designed and implemented first microbial time series observations of genetic diversity in the Arabian Gulf from scratch

Abu Dhabi, UAE
Aug 2018 - May 2019

LONG LAB

Research Assistant

- developed bubble traps to measure ebullition of oxygen bubbles from supersaturated seagrass meadows to constrain uncertainty in eddy covariance estimates of seagrass productivity
- interviewed and produced short documentary on the potential for seagrass meadows to act as long term carbon sink

Woods Hole, MA
May 2017 - Aug 2017

MINISTRY OF CLIMATE CHANGE

UNFCCC Student Delegate for the United Arab Emirates

- tracked negotiations on damage and loss at COP22
- informed UAE delegation on the scientific basis for climate change and ongoing oceanographic research on climate change in the Arabian Gulf and the Indian Ocean

Abu Dhabi, UAE
Aug 2016 - Dec 2016

PUBLICATIONS

- Robert Jarolim, Benoit Tremblay, Andrés Muñoz-Jaramillo, Kyriaki-Margarita Bintsi, Anna Jungbluth, Miraflor Santos, Angelos Vourlidas, James P. Mason, Sairam Sundaresan, Cooper Downs, and Ronald M. Caplan. SuNeRF: 3D Reconstruction of the Solar EUV Corona Using Neural Radiance Fields. *The Astrophysical Journal Letters*, 961(2):L31, January 2024. Publisher: The American Astronomical Society
- Kyriaki-Margarita Bintsi, Robert Jarolim, Benoit Tremblay, Miraflor Santos, Anna Jungbluth, James Paul Mason, Sairam Sundaresan, Angelos Vourlidas, Cooper Downs, Ronald M. Caplan, and Andrés Muñoz Jaramillo. SuNeRF: Validation of a 3D Global Reconstruction of the Solar Corona Using Simulated EUV Images, November 2022. arXiv:2211.14879 [astro-ph]

- Ahmed A. Shibl, Ashley Isaac, Michael A. Ochsenkühn, Anny Cárdenas, Cong Fei, Gregory Behringer, Marc Arnoux, Nizar Drou, Miraflor P. Santos, Kristin C. Gunsalus, Christian R. Voolstra, and Shady A. Amin. Diatom modulation of select bacteria through use of two unique secondary metabolites. *Proceedings of the National Academy of Sciences*, 117(44):27445–27455, November 2020

PRESENTATIONS

- Miraflor Santos. Using wavelet analysis to characterize seasonality in a phytoplankton community at Martha's Vineyard Coastal Observatory. *Association for the Sciences of Limnology and Oceanography Ocean Sciences Meeting*, June 2023
- Miraflor Santos. Aggregate and compositional variability of heterotrophic bacteria in nearshore waters of the Northeast U.S. Shelf. *Association for the Sciences of Limnology and Oceanography Ocean Sciences Meeting*, September 2022
- Miraflor Santos. Forecasting the dynamics of a marine microbial community at Martha's Vineyard Coastal Observatory (MVCO): a case study of Gaussian process regression modelling. *Ecological Forecasting Initiative Conference*, June 2022

SKILLS

- **Programming Languages:** Python, Julia, R, Matlab
- **Other Technologies:** Latex, HTML, CSS, Adobe Suite

LEADERSHIP AND HONORS

- Awarded **Ocean Ridge Initiative Fellowship** (First Year Graduate Fellowship) Sep 2019 - Sep 2020
- worked for Grad Students for a Healthy MIT to **increase mental health coverage** from 12 free sessions with 20 USD copay after to 52 sessions with 5 USD copay after for all graduate students Sep 2019 - Feb 2020
- Awarded **NYU Abu Dhabi Full Merit Scholarship** (70,000 USD per year scholarship) Aug 2014 - May 2019
- NYU Global Leaders in Change Fellow (won a **fellowship at a Global Leadership Conference in Shanghai, China** to learn about climate change policy and education in China) Jun 2017 - Aug 2017
- Won UNESCO-Salon Youth Video Competition and premiered winning film about **Education for Sustainable Development** at the Paris Climate Change Conference Jul 2015 - Dec 2015